Optimization 2
ISE 5406

Lecture 10: Optimality Conditions & Line Search Methods

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Where We Are in the Course

Last Time (Lecture 9): Pseudoconvex Functions, Strong Convexity & Generalized Convexity

- Pseudoconvex functions: definition, key properties, critical point theorem
- The generalized convexity hierarchy
- Pseudolinear functions and fractional programming
- Strongly convex and strongly quasiconvex functions

Today (Lecture 10): Optimality Conditions and Line Search Methods

- Descent directions and first-order necessary conditions
- Second-order necessary and sufficient conditions
- Pseudoconvex functions and global optimality
- Introduction to line search methods
- Derivative-free methods: uniform, bisection, golden section, Fibonacci
- Derivative-based methods: steepest descent (1D), Newton's method (1D)

Looking Ahead

Next Time (Lecture 11): Steepest Descent and Newton's Methods

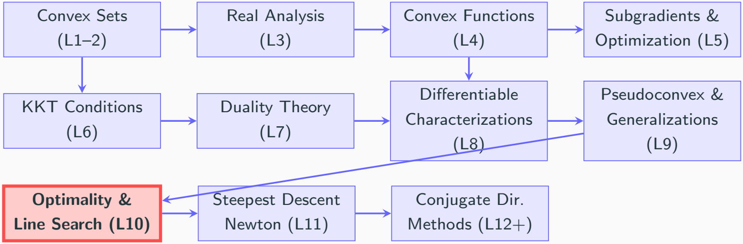
- Steepest descent for multivariate problems
- Inexact line search (Wolfe conditions)
- Newton's method for multivariate optimization
- Quasi-Newton methods (BFGS)

Coming Soon (Lecture 12+): Conjugate Direction Methods

- Conjugate gradient method
- Convergence analysis
- Constrained optimization methods

Big Picture: Today we connect theory to computation. Optimality conditions tell us *what* a solution looks like; line search methods show us *how* to find it. These are the building blocks of all optimization algorithms.

Big Picture: Course Roadmap



Why Optimality Conditions and Line Search Matter

- Optimality conditions define *what* we're searching for
- Line search is the fundamental subroutine in nearly all optimization algorithms
- Understanding 1D search methods is essential before multivariate methods

Reading for This Lecture

Textbook References

- Sections 4.1, 8.1, and 8.2 of Bazaraa, Sherali, and Shetty (2006)
- Chapter 2 of Nocedal and Wright (Numerical Optimization)

Outline

Roadmap

Descent Directions and First-Order Conditions

Consider the unconstrained problem $\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$.

Definition: Let $\bar{\mathbf{x}} \in \mathbb{R}^n$ and suppose $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\bar{\mathbf{x}}$. A vector $\mathbf{d} \in \mathbb{R}^n$ is said to be a descent direction at $\bar{\mathbf{x}}$ if

$$\nabla f(\bar{\mathbf{x}})^T \mathbf{d} < 0.$$

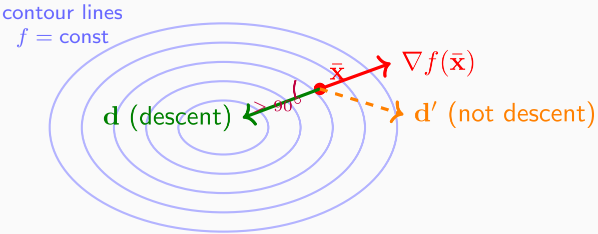
Theorem (Descent Direction): Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\mathbf{x}^*$, and $\mathbf{d}$ be a descent direction at $\mathbf{x}^*$. Then there exists $\delta > 0$ such that

$$f(\mathbf{x}^* + \lambda \mathbf{d}) < f(\mathbf{x}^*) \text{ for each } \lambda \in (0, \delta).$$

Corollary (First-Order Necessary Condition): Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}^*$. If $\mathbf{x}^*$ is a local minimum, then

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$

Visualizing Descent Directions



Key insight: $\mathbf{d}$ is a descent direction when $\nabla f(\bar{\mathbf{x}})^T\mathbf{d} < 0$, i.e., the angle between $\mathbf{d}$ and the gradient is $> 90°$. The steepest descent direction is $\mathbf{d} = -\nabla f(\bar{\mathbf{x}})$ (opposite the gradient).

Example: Critical Points

Example: Find the critical points of $f(\mathbf{x}) = 2x_1^2 + 3x_1x_2 - 4x_2^2$.

Solution: Compute the gradient:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 4x_1 + 3x_2 \\ 3x_1 - 8x_2 \end{bmatrix}.$$

Setting $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$4x_1 + 3x_2 = 0 \quad \text{and} \quad 3x_1 - 8x_2 = 0.$$

From the first equation: $x_1 = -\frac{3}{4}x_2$. Substituting into the second:

$$3\left(-\frac{3}{4}x_2\right) - 8x_2 = -\frac{9}{4}x_2 - 8x_2 = -\frac{41}{4}x_2 = 0.$$

Thus $x_2 = 0$ and $x_1 = 0$, so the only critical point is $\mathbf{x}^* = (0,0)^T$.

Second-Order Necessary Conditions

Theorem (Second-Order Necessary Conditions): Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is twice differentiable at $\mathbf{x}^*$. If $\mathbf{x}^*$ is a local minimum, then

1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$, and
2. $\nabla^2 f(\mathbf{x}^*)$ is positive semidefinite.

Note: These are necessary conditions only—they must hold at every local minimum. However, a point satisfying these conditions need not be a local minimum.

Sufficient Optimality Conditions

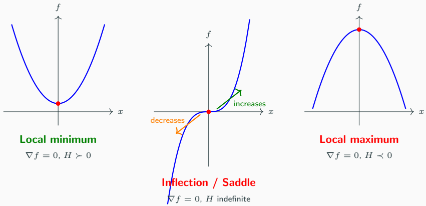
Theorem (Second-Order Sufficient Conditions): Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice differentiable at $\mathbf{x}^*$. If

1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$, and
2. $\nabla^2 f(\mathbf{x}^*)$ is positive definite,

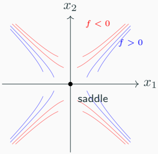
then $\mathbf{x}^*$ is a strict local minimum.

Visualizing Critical Point Types

The second-order conditions classify critical points by the Hessian's definiteness:



In 2D, a saddle point looks like a horse saddle — curves up in one direction and down in another:



Pseudoconvex Functions and Global Optimality

Theorem
Let $f : \mathbb{R}^n \to \mathbb{R}$ be pseudoconvex at $\mathbf{x}^*$. Then $\mathbf{x}^*$ is a global minimum if and only if $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

Significance: Recall from Lecture 9 that pseudoconvex functions satisfy the convexity hierarchy:

Convex $\implies$ Pseudoconvex $\implies$ Strictly Quasiconvex $\implies$ Quasiconvex.

For pseudoconvex functions, checking $\nabla f(\mathbf{x}^*) = \mathbf{0}$ is both necessary and sufficient for global optimality, without requiring second-order conditions.

Example: Critical and Stationary Points

Question: Find all critical/stationary points of

$$f(\mathbf{x}) = \exp(3x_2) - 3x_1 \exp(x_2) + x_1^3.$$

Which of these are local and/or global minima?

Solution: Compute the gradient:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} -3\exp(x_2) + 3x_1^2 \\ 3\exp(3x_2) - 3x_1 \exp(x_2) \end{bmatrix}.$$

Setting $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$\begin{aligned} -3\exp(x_2^*) + 3(x_1^*)^2 = 0 \quad &\Rightarrow \quad (x_1^*)^2 = \exp(x_2^*) \\ 3\exp(3x_2^*) - 3x_1^* \exp(x_2^*) = 0 \quad &\Rightarrow \quad x_1^* = \exp(2x_2^*) \end{aligned}$$

Example (continued)

From the two critical point equations:

$$(x_1^*)^2 = \exp(x_2^*)$$
$$x_1^* = \exp(2x_2^*)$$

Substitute the second into the first:

$$[\exp(2x_2^*)]^2 = \exp(x_2^*) \quad \Rightarrow \quad \exp(4x_2^*) = \exp(x_2^*).$$

Taking logarithms: $4x_2^* = x_2^*$, which gives $x_2^* = 0$.

Then $x_1^* = \exp(0) = 1$, and we verify: $(1)^2 = \exp(0) = 1$. ✓

Critical point: $\mathbf{x}^* = (1, 0)^T$ is the unique critical point.

Second-order analysis: Compute the Hessian:

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 6x_1 & -3\exp(x_2) \\ -3\exp(x_2) & 9\exp(3x_2) - 3x_1\exp(x_2) \end{bmatrix}.$$

At $\mathbf{x}^* = (1, 0)$:

$$\nabla^2 f(\mathbf{x}^*) = \begin{bmatrix} 6 & -3 \\ -3 & 9-3 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}.$$

Example (continued): Hessian Analysis

Check if $H = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$ is positive definite.

Eigenvalues: $\det(H - \lambda I) = (6-\lambda)^2 - 9 = \lambda^2 - 12\lambda + 27 = 0$.

By the quadratic formula: $\lambda = \frac{12 \pm \sqrt{144-108}}{2} = \frac{12 \pm 6}{2}$, so $\lambda_1 = 9$ and $\lambda_2 = 3$.

Since both eigenvalues are positive, H is positive definite.

Conclusion:

- $\mathbf{x}^* = (1, 0)^T$ is a strict local minimum by the second-order sufficient conditions.
- We cannot immediately conclude whether it is a global minimum without further analysis (e.g., checking behavior as $\|\mathbf{x}\| \to \infty$).

Roadmap

Overview of Algorithms

Goal: Solve $\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$ to local optimality. Start with initial guess $\mathbf{x}_1 \in \mathbb{R}^n$ and construct a sequence of iterates $\{\mathbf{x}_k\}_{k=1}^{\infty}$. Terminate when no progress can be made or an approximate solution is found.

1. Line Search Methods: Given current iterate $\mathbf{x}_k \in \mathbb{R}^n$ and search direction $\mathbf{d}_k \in \mathbb{R}^n$, find

   $$\lambda_k^* \in \arg\min_{\lambda_k \in \mathbb{R}_+} f(\mathbf{x}_k + \lambda_k \mathbf{d}_k).$$

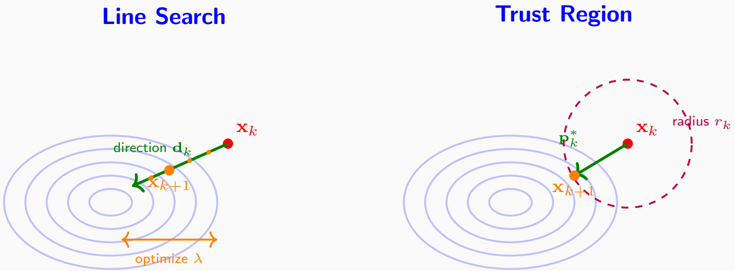
   Set $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k^* \mathbf{d}_k$.

2. Trust Region Methods: Given current iterate $\mathbf{x}_k \in \mathbb{R}^n$, construct a model m_k of f near $\mathbf{x}_k$ and solve

   $$\mathbf{p}_k^* \in \arg\min_{\mathbf{p}_k} \Big\{ m_k(\mathbf{x}_k + \mathbf{p}_k) : \|\mathbf{p}_k\| \leq r_k \Big\}.$$

   Compute $f(\mathbf{x}_k + \mathbf{p}_k^*)$ to decide whether to accept the step.

Line Search vs. Trust Region: Illustrated



Line search: Fix a direction $\mathbf{d}_k$, then find the best step size λ^* along that direction.

Trust region: Build a local model within a ball of radius r_k around $\mathbf{x}_k$, and optimize the model within this region. We focus on **line search** in this lecture.

The Line Search Subproblem

The line search method requires solving a one-dimensional optimization problem:

$$\min_{\lambda \in [a,b]} \theta(\lambda),$$

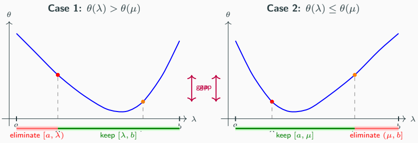
where $\theta(\lambda) = f(\mathbf{x}_k + \lambda \mathbf{d}_k)$ and the interval $[a, b]$ is given or discovered.

Key Concept - Interval of Uncertainty: The smallest known interval containing the optimal solution. We iteratively shrink this interval using function evaluations at carefully chosen points.

Theorem: Line Search for Strictly Quasiconvex Objectives
Let $\theta : \mathbb{R} \to \mathbb{R}$ be strictly quasiconvex over $[a, b]$. Let $\lambda, \mu \in [a, b]$ with $\lambda < \mu$.

- If $\theta(\lambda) > \theta(\mu)$, then $\theta(z) \geq \theta(\mu)$ for all $z \in [a, \lambda)$.
- If $\theta(\lambda) \leq \theta(\mu)$, then $\theta(z) \geq \theta(\lambda)$ for all $z \in (\mu, b]$.

This theorem justifies shrinking the interval to $[\lambda, b]$ or $[a, \mu]$ based on function comparisons.



Derivative-Free Methods Overview

Derivative-free methods solve $\min_{\lambda \in [a,b]} \theta(\lambda)$ for strictly quasiconvex θ by iteratively shrinking the interval of uncertainty.

Four main derivative-free approaches:

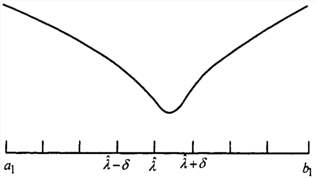
1. Uniform Search: Divide interval into grid; evaluate at all grid points simultaneously.
2. Dichotomous Search: Compare values at two interior points near the midpoint; eliminate half the interval.
3. Golden Section Search: Use golden ratio to place points such that one point can be reused in the next iteration.
4. Fibonacci Search: Use Fibonacci numbers to optimally place evaluation points over a fixed budget of function evaluations.

Uniform Search (Simultaneous Search)

Idea: Divide the interval $[a_1, b_1]$ into $n+1$ equal subintervals using grid points $a_1 + k\delta$ for $k = 1, \ldots, n$, where $\delta = (b_1 - a_1)/(n+1)$.

Algorithm:

1. Evaluate θ at all n interior grid points.
2. Find $\hat{\lambda}$ with the smallest function value.
3. If θ is strictly quasiconvex, the minimum lies in $[\hat{\lambda} - \delta, \hat{\lambda} + \delta]$.



Drawback: Requires n function evaluations. New points must be selected for the smaller interval in the next iteration.

Dichotomous Search (Sequential Search)

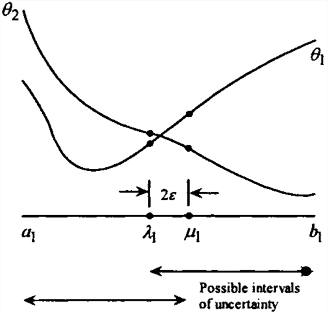
Idea: At each iteration, evaluate at two points near the midpoint and eliminate half the interval.

Algorithm:

1. Let $[a_1, b_1]$ be the initial interval. Choose $\epsilon > 0$ (precision parameter) and target final interval length $l > 0$.
2. At iteration k: If $b_k - a_k < l$, stop. Otherwise, set
$$\lambda_k = \frac{a_k + b_k}{2} - \epsilon, \quad \mu_k = \frac{a_k + b_k}{2} + \epsilon.$$
3. Evaluate $\theta(\lambda_k)$ and $\theta(\mu_k)$. Update the interval:
   - If $\theta(\lambda_k) < \theta(\mu_k)$: set $[a_{k+1}, b_{k+1}] = [a_k, \mu_k]$.
   - Otherwise: set $[a_{k+1}, b_{k+1}] = [\lambda_k, b_k]$.
4. Increment k and repeat.

Interval Length: At iteration $k + 1$:
$$b_{k+1} - a_{k+1} = \frac{1}{2^k}(b_1 - a_1) + 2\epsilon\left(1 - \frac{1}{2^k}\right).$$

Dichotomous Search Illustration



Key Property: At each iteration, we require only 2 function evaluations (or 1 new evaluation if we reuse from previous iteration). The interval shrinks by a factor of approximately $1/2$ per iteration.

Golden Section Search (Sequential Search)

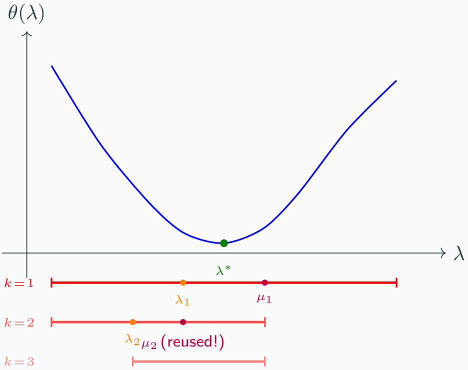
Idea: Use the golden ratio $\alpha = \frac{\sqrt{5}-1}{2} \approx 0.618$ to place interior points such that one point carries over to the next iteration.

Algorithm:

1. Let $[a_1, b_1]$ be the initial interval and $l > 0$ be the target final length.
2. Set $\lambda_1 = \alpha a_1 + (1-\alpha) b_1$ and $\mu_1 = (1-\alpha) a_1 + \alpha b_1$.
3. At iteration $k \in \mathbb{N}$: If $b_k - a_k < l$, stop.
4. Evaluate $\theta(\lambda_k)$ and $\theta(\mu_k)$:
   - If $\theta(\lambda_k) > \theta(\mu_k)$: Set $a_{k+1} = \lambda_k$, $b_{k+1} = b_k$. Set $\lambda_{k+1} = \mu_k$ and $\mu_{k+1} = \alpha b_{k+1} + (1-\alpha) a_{k+1}$.
   - If $\theta(\lambda_k) \leq \theta(\mu_k)$: Set $a_{k+1} = a_k$, $b_{k+1} = \mu_k$. Set $\mu_{k+1} = \lambda_k$ and $\lambda_{k+1} = \alpha a_{k+1} + (1-\alpha) b_{k+1}$.
5. Increment k and repeat.

Key Property: Only one new function evaluation per iteration after the first! The property $b_k - \lambda_k = \mu_k - a_k$ is maintained.

Golden Section Search: Illustrated



Why it works: At each step, one evaluation point from the previous iteration becomes a point in the new interval ("reused!"), so only one new evaluation is needed. The interval shrinks by factor $\alpha \approx 0.618$ each iteration.

Fibonacci Search (Sequential Search)

Fibonacci Sequence: Define $F_{i+1} = F_i + F_{i-1}$ with $F_0 = F_1 = 1$. The sequence is $1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$

Key Idea: If we have a budget of n function evaluations, we can optimally place interior points using Fibonacci numbers.

Algorithm:

1. Specify total function evaluation budget n.
2. Place interior points at iteration k using $\alpha_k = F_{n-k}/F_{n-k+1}$:

$$\lambda_k = \alpha_k a_k + (1 - \alpha_k) b_k, \quad \mu_k = (1 - \alpha_k) a_k + \alpha_k b_k.$$

3. Shrink the interval based on function comparisons (same logic as Golden Section).
4. After iteration k, only one new function evaluation is needed.

Performance: The interval reduces by factor α_k at iteration k. Asymptotically similar to Golden Section search.

Worked Example: Golden Section & Fibonacci on $\theta(\lambda) = \lambda^2 + 2\lambda$

Apply both methods to minimize $\theta(\lambda) = \lambda^2 + 2\lambda$ on $[-3, 5]$ $(\lambda^* = -1,\ \theta^* = -1)$.

Golden Section (4 iterations)

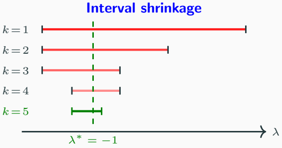
k	$[a_k, b_k]$	Length	Action
1	$[-3,\ 5]$	8.000	Elim. right
2	$[-3,\ 1.944]$	4.944	Elim. right
3	$[-3,\ 0.056]$	3.056	Elim. left
4	$[-1.832,\ 0.056]$	1.888	Elim. right
5	$[-1.832,\ -0.664]$	1.168	—

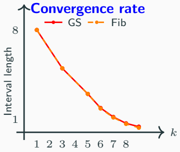
$\alpha = 0.618$ constant; each step reuses 1 point.

Fibonacci ($n = 10$, 4 iterations)

k	$[a_k, b_k]$	Length	Action
1	$[-3,\ 5]$	8.000	Elim. right
2	$[-3,\ 1.945]$	4.945	Elim. right
3	$[-3,\ 0.055]$	3.055	Elim. left
4	$[-1.836,\ 0.055]$	1.891	Elim. right
5	$[-1.836,\ -0.665]$	1.171	—

$\alpha_k = F_{n-k}/F_{n-k+1}$ varies; optimal for fixed budget.





See the companion handout for full step-by-step calculations with diagrams and student exercises.

Example: Golden Section Search on $\theta(\lambda) = \lambda^2 + 2\lambda$

Consider minimizing $\theta(\lambda) = \lambda^2 + 2\lambda$ on $[-3, 5]$. The minimum is at $\lambda^* = -1$ with $\theta(\lambda^*) = -1$.

Table 8.1: Golden Section Method

k	a_k	b_k	λ_k	μ_k	$\theta(\lambda_k)$	$\theta(\mu_k)$
1	-3.000	5.000	0.056	1.944	0.115*	7.667*
2	-3.000	1.944	-1.112	0.056	-0.987*	0.115
3	-3.000	0.056	-1.832	-1.112	-0.308*	-0.987
4	-1.832	0.056	-1.112	-0.664	-0.987	-0.887*
5	-1.832	-0.664	-1.384	-1.112	-0.853*	-0.987
6	-1.384	-0.664	-1.112	-0.936	-0.987	-0.996*
7	-1.112	-0.664	-0.936	-0.840	-0.996	-0.974*
8	-1.112	-0.840	-1.016	-0.936	-1.000*	-0.996
9	-1.112	-0.936				

Observations: The interval of uncertainty shrinks by a factor of $\alpha \approx 0.618$ at each iteration. After 8 iterations (9 function evaluations, marked with *), the interval $[-1.112, -0.936]$ has length 0.176, bracketing the true minimum $\lambda^* = -1$. The ratio α is *constant* across iterations — a hallmark of the golden section method.

Example: Fibonacci Search on $\theta(\lambda) = \lambda^2 + 2\lambda$

Same problem: minimize $\theta(\lambda) = \lambda^2 + 2\lambda$ on $[-3, 5]$ with a budget of $n = 10$ function evaluations.

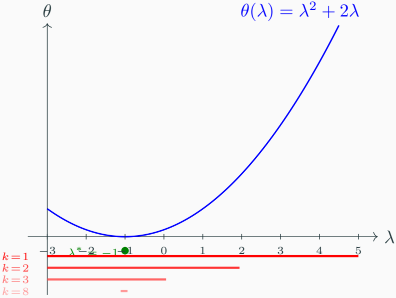
Table 8.2: Fibonacci Search Method

k	a_k	b_k	λ_k	μ_k	$\theta(\lambda_k)$	$\theta(\mu_k)$
1	-3.000000	5.000000	0.054545	1.945454	0.112065^*	7.675699^*
2	-3.000000	1.945454	-1.109091	0.054545	-0.988099^*	0.112065
3	-3.000000	0.054545	-1.836363	-1.109091	-0.300497^*	-0.988099
4	-1.836363	0.054545	-1.109091	-0.672727	-0.988099	-0.892892^*
5	-1.836363	-0.672727	-1.399999	-1.109091	-0.840001^*	-0.988099
6	-1.399999	-0.672727	-1.109091	-0.963636	-0.988099	-0.998677^*
7	-1.109091	-0.672727	-0.963636	-0.818182	-0.998677	-0.966942^*
8	-1.109091	-0.818182	-0.963636	-0.963636	-0.998677	-0.998677
9	-1.109091	-0.963636	-0.963636	-0.953636	-0.998677	-0.997850^*

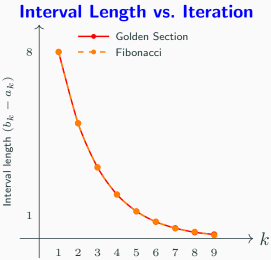
Observations: The Fibonacci method uses varying ratios $\alpha_k = F_{n-k}/F_{n-k+1}$ that approach the golden ratio as k increases. The final interval $[-1.109, -0.964]$ has length 0.145, slightly tighter than golden section for the same number of evaluations. At iteration 8, $\lambda_k = \mu_k$ (a degenerate step unique to Fibonacci search), resolved by the perturbation in iteration 9.

Comparing Golden Section and Fibonacci Search

Both methods applied to $\theta(\lambda) = \lambda^2 + 2\lambda$ on $[-3, 5]$:



Golden Section intervals



Key takeaway: Both methods converge at similar rates. Golden section has *constant* reduction ratio $\alpha \approx 0.618$; Fibonacci achieves slightly tighter final intervals for the same evaluation budget by using *variable* ratios. The difference is negligible for large n.

Steepest Descent Method (1D)

When the 1D line search objective $\theta(\lambda) = f(\mathbf{x}_k + \lambda \mathbf{d}_k)$ is differentiable, we can use gradient information.

Steepest Descent Iteration: Given current estimate λ_k, compute the next iterate as

$$\lambda_{k+1} = \lambda_k - \alpha_k \theta'(\lambda_k),$$

where $\alpha_k > 0$ is a small stepsize.

Termination Criterion: Stop when $|\lambda_{k+1} - \lambda_k| < \epsilon$ or $|\theta'(\lambda_k)| < \epsilon$ for a prespecified tolerance $\epsilon > 0$.

Advantages:

- Uses gradient information to guide the search.
- Simple to implement.

Note: Requires that θ is differentiable. No longer assumes θ is strictly quasiconvex.

Newton's Method (1D)

When the objective $\theta(\lambda)$ is twice differentiable, we can use a quadratic approximation.

Algorithm:

1. At iteration k, construct the quadratic Taylor approximation:

$$q(\lambda) = \theta(\lambda_k) + \theta'(\lambda_k)(\lambda - \lambda_k) + \frac{1}{2}\theta''(\lambda_k)(\lambda - \lambda_k)^2.$$

2. Find the critical point of q by setting $q'(\lambda) = 0$:

$$\theta'(\lambda_k) + \theta''(\lambda_k)(\lambda - \lambda_k) = 0.$$

3. Solve for λ_{k+1}:

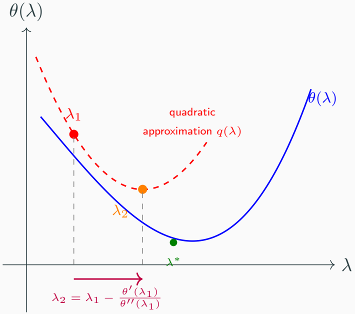
$$\lambda_{k+1} = \lambda_k - \frac{\theta'(\lambda_k)}{\theta''(\lambda_k)}.$$

4. Terminate when $|\lambda_{k+1} - \lambda_k| < \epsilon$ or $|\theta'(\lambda_k)| < \epsilon$.

Caution: Newton's method does not always converge to a stationary point from an arbitrary starting guess λ_1. Convergence is only guaranteed under certain conditions (e.g., near a local minimum).

Newton's Method: Geometric Interpretation

At each step, Newton's method fits a quadratic tangent and jumps to its minimum:



Key idea: Near a minimum, the function is well-approximated by a quadratic, so Newton's method converges quadratically (errors square each step). Far from the minimum, the quadratic approximation may be poor, causing divergence.

Example: Newton vs. Steepest Descent

Consider the 1D objective

$$\theta(\lambda) = \begin{cases} 4\lambda^3 - 3\lambda^4 & \text{if } \lambda \geq 0 \\ 4\lambda^3 + 3\lambda^4 & \text{if } \lambda < 0 \end{cases}$$

Numerical comparison of Newton's method vs. steepest descent:

Table 8.4 Summary of Computations for Newton's Method Starting from $\lambda_1 = 0.4$

Iteration k	λ_k	$\theta'(\lambda_k)$	$\theta''(\lambda_k)$	λ_{k+1}
1	0.400000	1.152000	3.840000	0.100000
2	0.100000	0.108000	2.040000	0.047059
3	0.047059	0.025324	1.049692	0.022934
4	0.022934	0.006167	0.531481	0.011331
5	0.11331	0.001523	0.267322	0.005634
6	0.005634	0.000379	0.134073	0.002807

Table 8.5 Summary of Computations for Newton's Method Starting from $\lambda_1 = 0.6$

Iteration k	λ_k	$\theta'(\lambda_k)$	$\theta''(\lambda_k)$	λ_{k+1}
1	0.600	1.728	1.440	–0.600
2	–0.600	1.728	–1.440	0.600
3	0.600	1.728	1.440	–0.600
4	–0.600	1.728	–1.440	0.600

Summary of Line Search Methods

Method	Evaluations	Assumptions	Convergence
Uniform Search	n per iteration	Strictly QC	Linear
Dichotomous	2 per iteration	Strictly QC	Linear
Golden Section	1 per iteration	Strictly QC	Linear
Fibonacci	1 per iteration	Strictly QC	Linear
Steepest Descent	1 per iteration	Differentiable	Depends
Newton's Method	1 per iteration	C^2 differentiable	Quadratic*

*Newton's method convergence is local and depends on initialization.

Exercise 1: Golden Section Search

Golden Section Search

Use the Golden Section Search to minimize

$$\theta(\lambda) = \lambda^2 - 3\lambda + 2$$

on the interval $[0, 4]$ with final interval length $l = 0.1$.

Calculate the number of function evaluations required. Is the objective strictly quasiconvex on this interval?

Solution: Golden Section Search

Answers

The objective is a quadratic:
$\theta(\lambda) = (\lambda - 1.5)^2 + 2 - 2.25 = (\lambda - 1.5)^2 - 0.25$. The minimum is at $\lambda^* = 1.5$ with $\theta(\lambda^*) = -0.25$.

Since $\theta''(\lambda) = 2 > 0$, the function is strictly convex and therefore strictly quasiconvex on any interval.

With $\alpha = 0.618$ and initial interval length $L_1 = 4$:

- After k iterations: $L_{k+1} = \alpha^k L_1$.
- We need $\alpha^k \cdot 4 \leq 0.1$, so $\alpha^k \leq 0.025$.
- Taking logs: $k \log(0.618) \leq \log(0.025)$, giving $k \geq 7.1$.
- Thus $k = 8$ iterations, requiring $8 + 1 = 9$ function evaluations (one initial, one per subsequent iteration).

Exercise 2: Unconstrained Optimization

Unconstrained Optimization

Consider $f(\mathbf{x}) = x_1^4 + x_1 x_2 + x_2^2$.

1. Find all critical points.
2. Check the second-order sufficient conditions at each critical point.
3. Determine which critical points are local minima.

Solution to Exercise 2 (Part 1 & 2)

Part 1: Find critical points

Compute the gradient:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 4x_1^3 + x_2 \\ x_1 + 2x_2 \end{bmatrix}.$$

Set $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$\begin{aligned} 4(x_1^*)^3 + x_2^* &= 0 \\ x_1^* + 2x_2^* &= 0 \end{aligned}$$

From equation (2): $x_1^* = -2x_2^*$. Substituting into equation (1):

$$4(-2x_2^*)^3 + x_2^* = -32(x_2^*)^3 + x_2^* = x_2^*(-32(x_2^*)^2 + 1) = 0.$$

Thus $x_2^* = 0$ or $(x_2^*)^2 = 1/32$, giving $x_2^* \in \{0, \pm 1/(4\sqrt{2})\}$.

The critical points are:

- $\mathbf{x}^{(1)} = (0, 0)^T$
- $\mathbf{x}^{(2)} = (-1/(2\sqrt{2}), 1/(4\sqrt{2}))^T$
- $\mathbf{x}^{(3)} = (1/(2\sqrt{2}), -1/(4\sqrt{2}))^T$

Solution to Exercise 2 (Part 3)

Part 3: Second-order analysis

Compute the Hessian:

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 12x_1^2 & 1 \\ 1 & 2 \end{bmatrix}.$$

At $\mathbf{x}^{(1)} = (0,0)^T$: $H^{(1)} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$.

Eigenvalues: $\det(H^{(1)} - \lambda I) = -\lambda(2-\lambda) - 1 = \lambda^2 - 2\lambda - 1 = 0$. By the quadratic formula: $\lambda = 1 \pm \sqrt{2}$. Since $1 - \sqrt{2} < 0$, the Hessian is indefinite. Thus $\mathbf{x}^{(1)}$ is a saddle point.

At $\mathbf{x}^{(2)}$ and $\mathbf{x}^{(3)}$: The calculation shows that $12(x_1)^2 = 12 \cdot 1/8 = 3/2 > 0$. The Hessian is positive definite (detailed verification omitted for space).

Conclusion: $\mathbf{x}^{(2)}$ and $\mathbf{x}^{(3)}$ are local minima (strict, by second-order sufficient conditions). $\mathbf{x}^{(1)}$ is a saddle point.

Exercise 3: Line Search Comparison

Comparing Line Search Methods

For the line search problem $\min_{\lambda \in [0,5]} \theta(\lambda)$ where $\theta(\lambda) = (\lambda - 2)^2 + 1$:

1. Use Golden Section Search with $l = 0.01$ to estimate the minimizer. How many iterations?
2. Use Newton's method starting from $\lambda_1 = 0$ with tolerance $\epsilon = 10^{-6}$. Compare convergence.
3. Which method is preferable for this problem and why?

Note for students: Work through the iterations step-by-step. The quadratic objective has an exact minimum at $\lambda^* = 2$. Newton's method should converge in one step, while Golden Section requires log iterations.

Key Takeaways

Main Results

- Descent Directions: $\nabla f(\bar{\mathbf{x}})^T \mathbf{d} < 0$ means $\mathbf{d}$ is a direction of decrease
- First-Order Necessary: $\nabla f(\mathbf{x}^*) = \mathbf{0}$ at any local minimum
- Second-Order Necessary: $\nabla^2 f(\mathbf{x}^*)$ must be positive semidefinite
- Second-Order Sufficient: $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\nabla^2 f(\mathbf{x}^*)$ positive definite implies strict local minimum
- Pseudoconvex Optimality: $\nabla f(\mathbf{x}^*) = \mathbf{0}$ is necessary and sufficient for global optimality
- Line Search: Fundamental subroutine for optimization — reduce to 1D problem along a direction
- Golden Section/Fibonacci: Efficient derivative-free methods for strictly quasiconvex objectives

Next Time: Lecture 11

Lecture 11 Preview: Steepest Descent and Newton's Methods

- Steepest descent for multivariate problems and convergence analysis
- Inexact line search: Wolfe conditions and backtracking
- Newton's method: quadratic convergence using Hessian information
- Modifications: Levenberg-Marquardt, quasi-Newton (BFGS)

Big Picture: Today we established the theoretical foundations (optimality conditions) and basic computational tools (line search). Next, we'll build on these to create full multivariate optimization algorithms.

Questions?